

OSBAMS Rev.2 controller PCB — test point map

The PCB has NO dedicated test points. Part A lists the existing pads that can be probed today; Part B is the requested list adapted to this board (items that need a layout change are marked NOT ON PCB — the design was not modified). Coordinates: mm from the lower-left board corner, Y up.

Part A — existing probe points

Signal	Best probe pad	Alternates	Purpose
+12V (after D3)	C1 pad 1 (67.6, 42.5) mm	D3 pad 2 (49.1, 51.4) mm, J3 pad 1 (25.1, 64.0) mm	supply verification (needs D3 correctly oriented)
12V_IN (before D3)	J1 pad 1 (61.8, 63.0) mm	D3 pad 1 (49.1, 66.6) mm	input present / reverse-polarity check
GND	J1 pad 2 (66.8, 63.0) mm	C1 pad 2 (65.6, 42.5) mm, J5 pad 2, any GND pad (B.Cu pour)	reference
+3V3 (from Nucleo)	J5 pad 1 (39.8, 15.0) mm	J4 pad 1 (64.1, 31.0) mm, U1 pad 5 (58.2, 23.0) mm, J6 pad 1	logic rail
SDA (I2C)	J4 pad 4 (71.7, 31.0) mm	J5 pad 3 (37.3, 15.0) mm, U1 pad 2 (53.1, 23.0) mm	TC74 / INA228 bus
SCL (I2C)	J4 pad 3 (69.2, 31.0) mm	J5 pad 4 (37.3, 12.5) mm, U1 pad 4 (56.5, 23.0) mm	TC74 / INA228 bus
UART_TX	J6 pad 3 (37.3, 24.5) mm	J5 pad 6 (34.8, 12.5) mm	debug / Pi link (3.3 V)
UART_RX	J6 pad 4 (39.8, 24.5) mm	J5 pad 7 (32.2, 15.0) mm	debug / Pi link (3.3 V)
GPIO_GATE (MCU output)	J5 pad 5 (34.8, 15.0) mm	R1 pad 1 (20.2, 19.5) mm	relay-driver command
GATE (MOSFET gate)	R1 pad 2 (10.0, 19.5) mm	Q1 pad 1 (11.0, 26.5) mm, R2 pad 1	gate drive after R1
COIL_SW (MOSFET drain)	Q1 pad 2 (13.5, 26.5) mm	J2 pad 2 (16.0, 64.0) mm, D2 pad 1	coil switch node
COIL_V (coil supply)	J2 pad 1 (11.0, 64.0) mm	J3 pad 2 (30.1, 64.0) mm, D2 pad 2	E-stop-gated coil supply

Part B — requested test points adapted to this board

TP	Signal	Purpose	Status on this PCB
TP1	12V	supply verification	no TP — probe C1.1 / D3.2 (Part A)
TP2	5V	logic rail	NOT ON PCB — there is no 5 V rail
TP3	3.3V	MCU rail	no TP — probe J5.1 (3.3 V comes from the Nucleo)
TP4	GND	reference	no TP — any GND pad / pour
TP5	INA228 SDA	communication	no TP — J4 pad 4 (module header)
TP6	INA228 SCL	communication	no TP — J4 pad 3
TP7	ADC voltage	calibration	NOT ON PCB — there is no ADC divider
TP8	UART TX	debug	no TP — J6 pad 3
TP9	UART RX	debug	no TP — J6 pad 4
TP10	E-stop	safety state	NOT ON PCB — J3 is a series loop; there is no E-stop sense signal
(new)	COIL_SW / GATE	relay-driver check	no TP — Q1 pads (Part A)

Recommendation (design change, not done): add labelled 1 mm test pads for +12V, +3V3, GND, SDA, SCL, GATE, COIL_SW before a production order.